Analyzing Retirement Portfolios with an Economic Scenario Generator

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Abstract

Effective retirement and financial planning requires tools to model and simulate future financial markets. We build an economic scenario generator (ESG) to simulate these potential future markets. Components of the ESG include inflation, interest rates, equity returns, and bond returns. This ESG provides the tools needed to analyze longevity risk for specific financial situations and, in turn, helps answer questions such as "how much should be annuitized?" and "what level of spending is sustainable?"

Keywords: economic scenario generator, financial planning, annuity, longevity risk, scenario simulation

1 Introduction

Financial and retirement plans are constructed based on a model of the future economy. The effectiveness of a specific plan can be evaluated by how well it performs given the model of the future economy. It follows then that the quality of these financial plan evaluations depends mostly on the quality of the model being used to represent states of the future economy. A poor model of the future will not yield reliable results regarding financial plan evaluations.

Currently, the financial planners mainly use deterministic techniques to model the future economy. Economic factors such as inflation, equity returns, and interest rates are given a fixed value. A popular approach for determining these values is to take averages from the past. These fixed values are then applied to a financial plan and the success of the plan can be evaluated.

A more thorough technique for future modeling includes directly using economic variable paths from the past. In this technique, the model for the future is fully determined by real values from the past economy. The quality of a financial plan is determined by then "applying" the plan to a specific year from the past and evaluating how well it would have performed given the past economic factors. This process can then be iterated using different starting points, representing different cohorts. This technique tells us exactly how a financial plan would have performed in the past, and it may be reasonable to posit that this gives planners a good understanding of how the plan will perform in the future.

However, both of these approaches are deterministic and, by nature, ignore the uncertainty and volatility of future financial markets. To better understand the performance of a financial plan, we propose a stochastic process through the use of an economic scenario generator (ESG). ESGs simulate future economic scenarios by modeling, jointly, economic variables and financial market values. ESGs are widely used in the financial and actuarial world to price financial products and quantify the risk associated with financial markets. We extend the application of ESGs to the personal financial planning sector and develop an ESG specifically for the purpose of evaluating personal financial plans. With an ESG that properly models underlying economic variables and their uncertainty, we can better quantify the risk associated with a given financial plan.

The primary risk we seek to quantify is longevity risk, or the probability that an individual outlives their assets. We want to explore what kinds of decisions can be made to reduce longevity risk. In a financial plan, there are three main "knobs" that can be adjusted: income, expenses, and asset allocation. Adjusting these levers allows us to address core planning questions such as: What level of spending is sustainable? How much should I save before retirement? Given my current financial position, how much should I annuitize to reduce longevity risk? While these are fundamental retirement planning concerns, an ESG allows us to evaluate them with greater depth by explicitly accounting for economic uncertainty.

The rest of the report is organized as follows. Section 2 is a literature review covering financial planning and portfolio construction, the question of how much to annuitize, and the ESG. Section 3 introduces the overall structure of the proposed ESG and explains each piece of the model of the economy. In section 4, simulations are applied to financial plans involving variable withdrawal rates and annuitization levels and the results are explored. Section 5 concludes and provides suggestions for further research in the area.

2 Literature Review

2.1 Financial Planning and Portfolio Construction

This section explores the relevant research done in economic variable modeling in the context of financial planning. Some ground has been made in exploring what types of models would be good engines in exploring financial plans. Because economic variables exhibit time-dependent behavior, they are most naturally modeled using time-series methods. Pang and Warshawsky (2010) use a vector autogressive (VAR) model to model jointly equity, bond, and cash returns, and search for “good wealth distribution strategies in retirement” using stochastic draws from this model. VAR models are widely used for capturing the time-dependent relationships among multiple variables. By allowing each variable to depend on its own past values as well as the past values of other variables, VAR models can represent well the co-movement of certain economic variables. Campbell and Viceira (2005) also use a VAR model to model short-term interest rates, dividend yield, and the slope of the yield curve. Estimated coefficients and covariances between variables are then used to analyze the annualized risk underlying the assets connected to these variables (T-bills, equity returns, etc.).

Collins et al. (2015) show that having an oversimplistic inflation model can lead to drastically different retirement income risk results. Just using different (but plausible) deterministic inflation measures yields retirement portfolio success metrics ranging from 27% to 57%, per their success criteria. They also explore different asset returns models by modeling jointly a two-asset class portfolio under regime-switching dynamics and show that sustainability rates change drastically when paths of economic variables driven by stochastic models are used in portfolio simulation.

2.2 How Much to Annuitize

Where money should be invested in retirement is perhaps the most important question a retirement planner needs to address, and as such, there is a plethora of commentary and research on the subject. Life-time annuitization is a popular choice because of the guaranteed income it provides in retirement. A guaranteed income up until death can be a great asset, depending on the retirement goals of the retiree. Yaari (1965) explains simply that if a consumer has no motive to leave a bequest after death, they should invest all of their assets in annuities. Annuities typically have a lower rate of return than equities, so investing only in equities historically allows you to build wealth while still consuming at the same rate at which an annuity would provide. Below-average equity returns, however, could introduce more risk to a portfolio. Poor timing of a market downturn could result in financial ruin, while a fixed-income investment such as an annuity can protect from that risk.

Milevsky (1998) takes an interesting approach to the question of whether or not to annuitize. He finds that a female retiree at 65 has a 95% chance of beating the mortality adjusted rate of return for a life-annuity by investing in equities up until age 80. He then proposes that the annuitant wait until their chances of beating the annuity's mortality adjusted rate of return are low enough to justify the purchase of an annuity.

Similar to Yaari (1965), Pang and Warshawsky (2009) find that life-annuities ignore considerations of bequest and distribute the highest guaranteed life income to the retiree. They find further in Pang and Warshawsky (2010) that an optimal annuitization strategy for a retiree at 65 is to begin by annuitizing 10% of wealth initially and then gradually annuitizing more and more up to 100% over 20 years.

2.3 ESGs

One of the earliest and most influential ESGs in the literature is provided by Wilkie (1986). His model adopts a cascade-style structure where the price index drives an equity index, which drives equity dividends and bond yields. Wilkie (1995) further extends this framework to include a wage index and short-term interest rates.

There is a wide variety of models that have been used in further literature to build the ESG, extending from Wilkie (1986). Wilkie (1986) primarily employs autoregressive dynamics in his cascade-structure ESG. Ahlgrim et al. (2005) employ regime-switching dynamics in the modeling of some of the economic factors in their ESG. Bégin (2021) explores the use of generalized autoregressive conditional heteroskedasticity (GARCH) with regime-switching dynamics using the Bayesian paradigm. Bégin (2021) further explores the performance of his proposed ESG with previously proposed ESGs and finds that from an in-sample performance perspective a more complex ESG has better performance, but evaluating based on out-of-sample metrics may point to different results. Bégin (2023) goes even further to propose an ensemble ESG that uses model weights for eight different ESGs and finds that this ensemble ESG creates reasonable and robust results even when applied to different countries' economies.

ESGs have been primarily proposed and built for the pricing of complex financial derivatives or insurance products. They have also been used to quantify business risk and have focused on institutional applications. However, relatively little attention has been given to using an ESG in assessing the risk inherent in an individual financial plan, leaving a notable gap in the literature.

3 Economic Scenario Generator

Many different ESG structures have been proposed (Bégin (2021), Bégin (2023), Wilkie (1995)). Our approach borrows heavily from existing models for the individual components, and while it does not introduce novel modeling techniques, it is uniquely structured to address the specific financial planning questions we aim to answer. In our ESG, we model inflation, interest rates, equity returns, fixed-income returns, and mortality rates. To account for dependencies between these variables, we adopt a cascade-style structure (Pedersen et al. (2016)). This structure is illustrated in Figure 3.1 below.

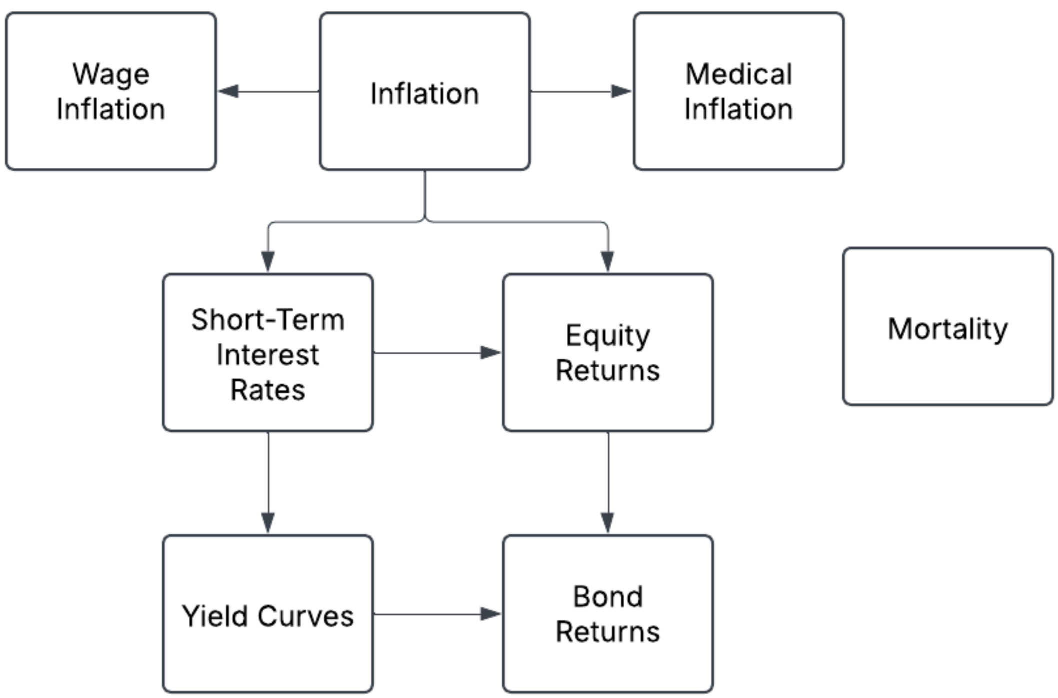

Figure 3.1: Cascade-style structure of the Economic Scenario Generator.

It is important to note that in this paper we don't incorporate wage inflation or bond returns in any financial simulations generated by the ESG. We build a model for wage inflation to be used in further work. Bond returns will also be used in extensions from this paper to explore more complex investment portfolios.

3.1 Model Selection Criteria

As previously alluded to, models for the ESG were primarily based on models used in existing ESG literature. For our use case, the most important factors in model selection (ordered by relative importance) are:

1. Behavior of simulated draws
2. Model fit
3. Model parsimony

As described in Bégin (2021), from an out-of-sample perspective, simpler models underlying an ESG perform better or similar to complex methods. Because of this, we try to keep our models as parsimonious as possible while still maintaining good model fit and reasonable simulated draws.

3.2 Data

We use United States economic data when fitting these models. Additionally, we estimate the ESG only using data from 1990 onward. This choice balances the need for a sufficient sample size while remaining representative of modern economic variable dynamics and encompassing a wide range of economic conditions, including market crashes, recessions, and expansions.

Inflation and interest rate data are obtained from the Federal Reserve Economic Data (FRED) database maintained by the Federal Reserve Bank of St. Louis (2025). Consumer price index (CPI), medical CPI, and interest rates are all recorded monthly, while the employment cost index (ECI) is recorded quarterly. Equity returns are modeled from the S&P 500 Total Return Index obtained from the tidyquant R package (Dancho and Vaughan

(2025)). For mortality, we use a 2012 Private Retirement Plans (PRI) mortality table for a male retiree provided by the Society of Actuaries (2019).

3.3 Inflation Models

The base level of our cascade-structure ESG is inflation. Unique to our use case, we are interested in modeling general inflation, medical inflation, and wage inflation separately, so we model general inflation excluding medical goods.

3.3.1 General Inflation

We measure general inflation with the CPI without medical goods. Let CPI_t denote the index at time t. Because CPI is non-stationary and generally increases over time, we model changes in CPI rather than its level. Specifically, we consider the log-ratio:

$$q_t = \log\left(\frac{\mathrm{CPI}_t}{\mathrm{CPI}_{t-1}}\right).$$

We assume that q_t follows an ARMA(1,1) process with constant mean, defined as:

$$q_t = \mu^{(q)} + \phi^{(q)} q_{t-1} + \theta^{(q)} \epsilon_{t-1}^{(q)} + \epsilon_t^{(q)}, \quad \text{where } \epsilon_t^{(q)} \sim N(0, \sigma^{2(q)}).$$

Here $\phi^{(q)}$ represents the auto-regressive component and $\theta^{(q)}$ represents the moving average component, both of order one.

3.3.2 Medical Inflation

We measure medical inflation using a consumer price index for just medical care. Let MedCPI_t denote the index at time t. The same way we modeled general inflation, we model medical inflation as a log-ratio:

$$m_t = \log\left(\frac{\mathrm{MedCPI}_t}{\mathrm{MedCPI}_{t-1}}\right),$$

where m_t follows an ARMA(1,1) process with a constant mean.

The original ESG structure proposed that medical inflation should depend on general inflation (q_t). We didn't find enough dependence between these two variables so we don't include general inflation in the model.

3.3.3 Wage Inflation

We measure wage inflation using the ECI. Let ECI_t denote the index at time t. Similar to the previous two inflation measures, we take the log-ratio of wage inflation:

$$w_t = \log\left(\frac{\text{ECI}_t}{\text{ECI}_{t-1}}\right),$$

where w_t follows an AR(1) process with a constant mean, and, per our cascading structure, general inflation (q) is included as an external regressor:

$$w_t = \mu^{(w)} + \gamma^{(w)} q_t + \phi^{(w)} w_{t-1} + \epsilon_t^{(w)}, \quad \text{where } \epsilon_t^{(w)} \sim N(0, \sigma^{2(w)}).$$

It is important to note that, although ECI data are provided at a quarterly frequency by the FRED, we choose to only model annual ECI. We found that quarterly ECI showed extremely strong negative autocorrelation, influencing model fit in a way that was not productive for our use case. To simplify, we model annual ECI and then linearly interpolate monthly values when obtaining simulated draws from the fitted model.

3.4 Interest Rates

The next piece of our ESG is interest rates. Perhaps one of the most crucial pieces of our ESG, interest rates directly affect the price and returns of certain fixed-income assets and indirectly affect the behavior of the prices and returns of other assets, most importantly equity returns. We are interested in modeling interest rates of varying duration to properly price fixed income assets such as annuities. As such, we model the yield curve by breaking it up into three components: the level, slope, and curvature. It has been found that these three factors account for about 99% of the variation in the yield curve (Litterman and Scheinkman, 1991), and as such, these three components provide a reasonable representation of the yield curve.

We denote interest rates as $\tilde{r}_{i,t}$, where $\tilde{r}_{i,t}$ is the i-year interest rate at time t (e.g., $\tilde{r}_{10,1}$ is the 10-year interest rate at time 1). We collect and use data for $i = \{3/12, 1, 2, 3, 5, 7, 10, 20, 30\}$. Here, 3/12 denotes the 3-month interest rate, and all other interest rates are yearly integer values.

When modeling interest rates, we must account for the fact that, in the United States, they effectively have a zero lower bound. This lower bound introduces skewness into the data, violating the assumption of normally distributed errors in standard time-series models when interest rates get close to zero. To account for this, we use a simplified version of a piecewise transformation proposed by Bégin (2021):

$$\tilde{r}_{i,t} = \begin{cases} r_{i,t} & \text{if } r_{i,t} > \overline{r} \\ \overline{r} - \overline{r}\log(\overline{r}) + \overline{r}log(r_{i,t}) & \text{if } r_{i,t} \leq \overline{r} \end{cases},$$

where $r_{i,t}$ is the rate at time t and $\overline{r}$ is the threshold at which we apply the log transformation. For our modeling we choose to set $\overline{r} = 0.005$, thus applying the transformation to all interest rates below 0.5%.

3.4.1 Short-Term Interest Rate

In theory, the level of a yield curve can be determined by any interest rate, but we decide to represent it by the short-term interest rate, specifically the three-month rate. Per our cascading ESG structure, we also incorporate short-term rates in our equity return model. In this way, a model specifically for short rates will prove beneficial for both the yield curve and the equity return pieces of our ESG.

Short-term interest rates have been found to exhibit conditional heteroskedasticity, that is to say, the variance of these interest rates is not just non-constant, but also can depend on the variance of the values before them. For example, large shocks in the interest rates tend to be followed by periods of high variance. To capture these dynamics, we choose to model the variance of the rates as a GARCH(1,1) process. The mean, like inflation, is modeled as an ARMA(1,1) process and, per our cascading structure, CPI is included as a regressor as $q_t = \log\left(\frac{\text{CPI}_t}{\text{CPI}_{t-1}}\right)$. More specifically:

$$\tilde{r}_{3/12} = \mu^{(\tilde{r}_{3/12})} + \gamma^{(\tilde{r}_{3/12})} q_t + \phi^{(\tilde{r}_{3/12})} \tilde{r}_{t-1} + \theta^{(\tilde{r}_{3/12})} \epsilon_{t-1}^{(\tilde{r}_{3/12})} + \epsilon_t^{(\tilde{r}_{3/12})}, \quad \text{where } \epsilon_t^{(\tilde{r}_{3/12})} \sim \mathcal{N}(0, \sigma_t^{2(\tilde{r}_{3/12})}),$$

and

$$\sigma_t^{2(\tilde{r}_{3/12})} = \omega^{(\tilde{r}_{3/12})} + \alpha^{(\tilde{r}_{3/12})} \epsilon_{t-1}^{2(\tilde{r}_{3/12})} + \beta^{(\tilde{r}_{3/12})} \sigma_{t-1}^{2(\tilde{r}_{3/12})},$$

for which $\alpha^{(\tilde{r}_{3/12})}$ is the reaction (or ARCH) component and $\beta^{(\tilde{r}_{3/12})}$ is the persistence (or GARCH) component of the variance model.

3.4.2 Yield Curve

We approximate the functional form of the yield curve using its level, slope, and curvature. Any point on the yield curve can then be approximated as a weighted sum of the level, slope, and curvature.

We define the level, slope, and curvature specifically in the following way:

$$\begin{aligned}\text{Level} &: \tilde{r}_{3/12,t}\\ \text{Slope} &: \tilde{r}_{30,t} - \tilde{r}_{3/12,t},\\ \text{Curvature} &: \tilde{r}_{3/12,t} + \tilde{r}_{30,t} - 2\tilde{r}_{10,t}.\end{aligned}$$

As described above, we model the level with a GARCH process. We choose to model slope and curvature as a two-dimensional VAR process of order one as follows:

$$\boldsymbol{F}_t = \boldsymbol{\mu} + \boldsymbol{A}\,\boldsymbol{F}_{t-1} + \boldsymbol{b}\,\tilde{r}_{3/12,t} + \boldsymbol{\nu}_t, \quad \text{where } \boldsymbol{\nu}_t \sim N_2(\boldsymbol{0}, \boldsymbol{\Sigma}),$$

where $\boldsymbol{F}_t$ is a 2×1 vector of slope and curvature at time t, $\boldsymbol{A}$ is a 2×2 matrix of auto-regressive components, $\boldsymbol{b}$ is a 2×1 vector of coefficients capturing the influence of the short-term rate at time t on each component, and $\boldsymbol{\Sigma}$ is the 2×2 variance-covariance matrix. We include the 3-month rate as a regressor to capture the dependence of slope and curvature on the overall level of the yield curve. Empirically, including this term also improves the fit of the VAR model.

We then model each interest rate as a linear function of level, slope, and curvature as follows:

$$\tilde{r}_i = \boldsymbol{x}'\boldsymbol{\beta}_i + \epsilon_i, \quad i = \{1, 2, 3, 5, 7, 20\}, \quad \text{where } \epsilon_i \sim N(0, \sigma_i^2).$$

Here, $\boldsymbol{x}$ is a vector of level, slope, and curvature, and $\boldsymbol{\beta}_i$ is a 3×1 vector of the weightings for interest rate i. Noticeably, we exclude the three-month, 10-year, and 30-year rates from this step because these rates are direct functions of level, slope, and curvature as described above.

In combination, the GARCH model for level, VAR model for slope and curvature, and linear regressions for each interest rate provide a complete framework to approximate the yield curve across all maturities. To illustrate the result of these models, figure 3.2 shows a simulated yield curve obtained from sampling a level, slope, and curvature value and then applying them to the fitted weightings for each interest rate (along with a smoothed line to approximate the curve).

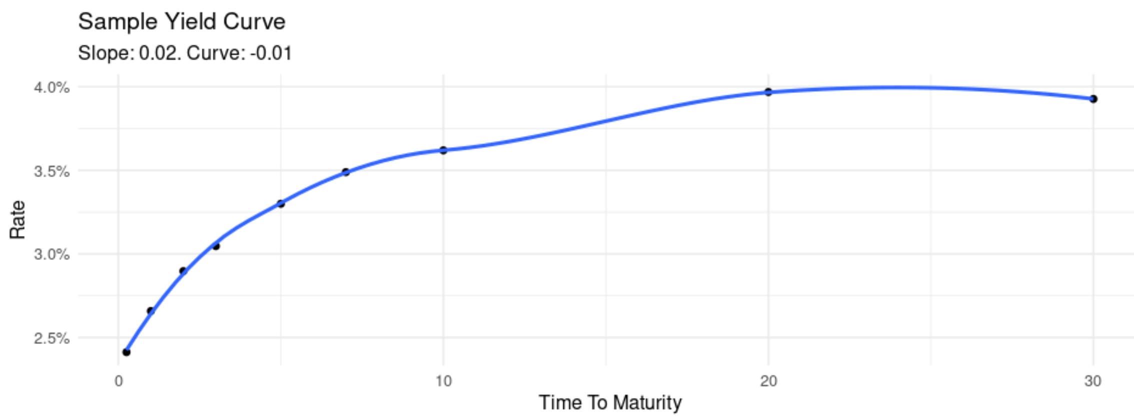

Figure 3.2: Sampled yield curve from Short Term GARCH model, VAR(1), and factor loadings for each interest rate $\tilde{r}_i$

3.5 Equity Returns

Equity returns is the last piece of our proposed ESG. Along with GARCH models, regime-switching models are commonly used to model equity returns. Regime-switching, otherwise known as Markov-switching models, assume that a series belongs to N regimes that are either observed or unobserved. In the case of two regimes, from time t to time $t+1$ there is some probability that the regime stays the same, or switches to the other regime. The parameters of interest change depending on the regime the random variable is assigned to.

In our case, we assume that the regimes underlying equity returns are unobserved. Because they are unobserved, we must determine the number of regimes to estimate and then estimate the underlying regimes the observations belong to. Existing literature has suggested that a two-regime model provides the best, parsimonious fit for equity returns (Hardy, 2001), so we assume there are two underlying regimes in the series.

Similar to inflation, we model the log-ratio of returns: $y_t = \log\left(\frac{S_t}{S_{t-1}}\right)$, where S_t is the total returns at time t. We model the mean of y_t without regime-switching dynamics and model the variance of y_t as a regime switching GARCH process as follows:

$$y_t = \mu^{(y)} + \gamma_1^{(y)} q_t + \gamma_2^{(y)} r_{3/12,t} + \epsilon_t^{(y)}, \quad \text{where } \epsilon_t^{(y)} \sim \mathcal{N}(0, \sigma_t^{2(y)}),$$

and

$$\sigma_t^{2(y)} = \omega_{R_t}^{(y)} + \alpha_{R_t}^{(y)} \epsilon_{t-1}^{2(y)} + \beta_{R_t}^{(y)} \sigma_{t-1}^{2(y)}, \quad \Theta = \begin{pmatrix} \theta_{11} & \theta_{12} \\ \theta_{21} & \theta_{22} \end{pmatrix}$$

for which R_t is the regime at time t. Θ represents the transition matrix, where each element θ_{ij} gives the probability of transitioning to regime j at time $t+1$ given that the process is in regime i at time t (e.g., θ_{12} is the probability of moving from regime 1 at time t to regime 2 at time $t+1$). The model for $\sigma^{2(y)}$ therefore has 6 fitted parameters (i.e., $\omega_1^{(y)}, \omega_2^{(y)}, \alpha_1^{(y)}, \alpha_2^{(y)}, \beta_1^{(y)}, \beta_2^{(y)}$). Per the ESG cascading structure, inflation (q_t), and the three-month interest rate ($r_{3/12,t}$) are both included as predictors in the mean model of y_t.

4 Case Study

After fitting the ESG, we can use it as the engine to answer a wide range of financial planning questions. As stated previously, perhaps the most fundamental question in financial planning is "how much can a retiree safely spend without exhausting their assets before death?" This question relates to longevity risk, defined as the risk of outliving one's assets. Accordingly, a primary objective of retirement planning is to minimize longevity risk while sustaining a level of spending deemed reasonable by the retiree. In an effort to decrease the risk of outliving your assets, you can purchase a lifetime annuity. A lifetime annuity is a financial tool that guarantees a constant stream of payments up until death to the retiree, and therefore, in theory, can decrease a retiree's longevity risk. Traditionally, this is purchased at retirement. With our ESG, we can extend the simple scenario from assessing longevity risk for a given level of spending to determining the optimal level of annuitization that minimizes longevity risk given the same level of spending.

4.1 Setup

To address this question, we produce simulated draws from the ESG of the different economic variables and apply those simulations to a financial plan, varying the level of spending and annuitization level. Here is an outline of this process:

1. We generate 10,000 simulations of the inflation, the interest rates, the yield curves, and the equity returns 50 years into the future.

2. We generate 10,000 sample death ages using the mortality table.

3. We apply these simulations to the financial plan, given a level of spending and annuitization.

4. We calculate success rate.

For this case study, we define success as dying before the portfolio value hits zero. The probability of success is therefore the proportion of simulations that satisfy this condition. This case study has two variables: the annuity proportion and the withdrawal rate. The annuity proportion, α, is simply the proportion of savings that is invested in a lifetime annuity. The rest of the portfolio is invested in the stock market.

4.1.1 Annuity Pricing

Before simulating a portfolio, we first specify how annuity income is determined. The base annuity price at age x, p_x is calculated the following way:

$$p_x = (1 + l) \sum_{t=1}^{T} (1 + r_{t/12})^{-t/12} \kappa_{x+\frac{t}{12}},$$

where $\kappa_{x+\frac{t}{12}}$ is the survival probability at age $x + \frac{t}{12}$ (in years), l is some insurer load, and $r_{t/12}$ is the three-month interest rate at time t. In our ESG, we model certain interest rates directly, but for interest rates not modeled directly (i.e., r_i such that $i \notin \left\{\frac{3}{12}, 1, 2, 3, 5, 7, 10, 20, 30\right\}$), we linearly interpolate the rate. For the terms below three months or above 30 years, we set the interest rate equal to the three-month rate and the 30-year rate, respectively. With the price p_x and the annuity proportion α we calculate the monthly annuity payout b as:

$$b = \frac{\alpha v_1}{p_x},$$

where v_1 is the portfolio starting value.

4.1.2 Withdrawal Rates

A withdrawal rate is defined as:

$$r = \frac{e_1}{v_1}$$

where e_1 is the yearly withdrawal amount at the beginning of the simulation (this can be thought of as yearly expenses at the time of retirement), and v_1 is the portfolio value at the beginning of the simulation. The beginning of the simulation in this case is the start of retirement. For a given retiree, we can determine their monthly expenses given their withdrawal rate as:

$$c = v_1 \frac{r}{12}.$$

4.1.3 Portfolio Value

With monthly expenses and a monthly annuity payout, we can now calculate how much should be drawn out of the stock market to cover monthly expenses (adjusting for inflation) as:

$$\mathrm{CPI}_t c - b.$$

With these pieces in place, we can simulate a retirement portfolio in the following manner:

$$v_t = (v_{t-1} - (\mathrm{CPI}_t c - b))y_t,$$

where y_t is the equity returns at time t, and c and b are direct functions of the withdrawal rate and the annuity proportion, and thus change as we change those values. It is important to note that with this structure, any excess money from an annuity payout past expenses (i.e. $\mathrm{CPI}_t c - b < 0$) gets invested into the stock market. Because of this structure, there may be scenarios where 100% annuitization can still result in above 0 portfolio values.

One final addition we make to our simulation is medical inflation. According to Butrica et al. (2005), typical married retirees spend about 20% of their monthly expenditures on health care. We take this finding and adjust our simulation in the following way:

$$v_t = (v_{t-1} - ((0.8 \times \mathrm{CPI}_t + 0.2 \times \mathrm{MedCPI}_t)c - b))y_t$$

where the monthly expenses, c, are adjusted proportionally for both medical inflation and general inflation.

4.2 Results

We conducted 10,000 simulations over a 50-year horizon for a 60-year-old male retiree. Each simulation was applied to a financial plan evaluated over a grid of withdrawal rates from 1% to 20% and annuity proportions ranging from 0% to 100% with 10% increments.

Importantly, we assume that a retiree maintains the same level of spending throughout retirement (i.e., c stays constant through the simulation). Figure 4.1 reports the success rate for each withdrawal rate and annuity proportion.

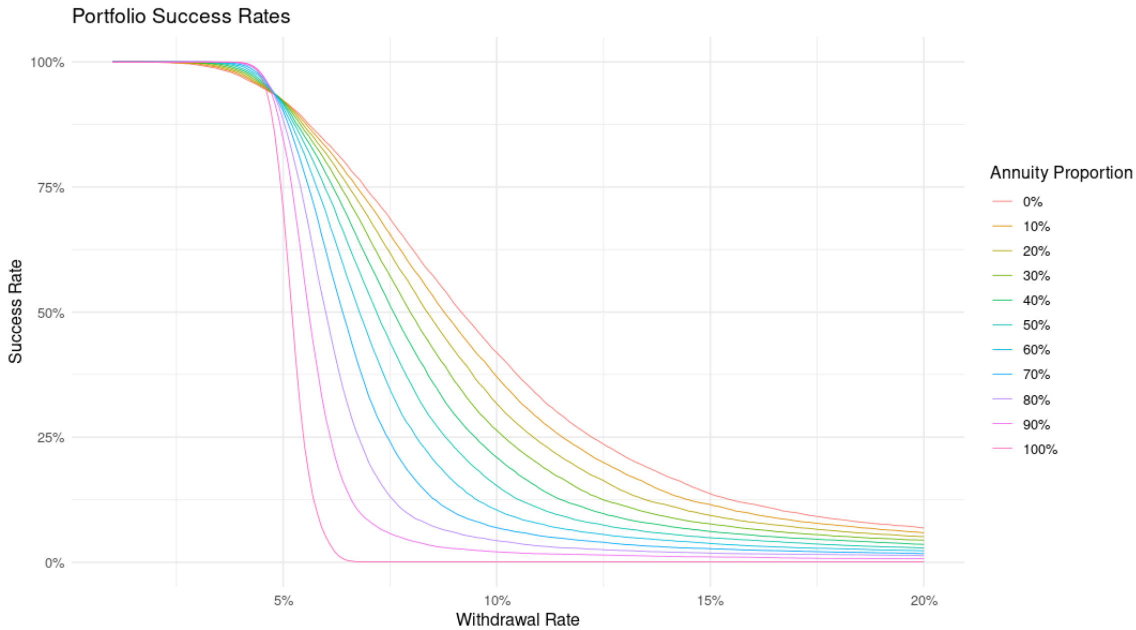

Figure 4.1: Portfolio Success Rates For Different Withdrawal Rates and Annuitization Levels

As expected, for all levels of annuitization, an increase in withdrawal rate results in a decrease in the probability of success. The rate at which the probability of success decreases changes, however, for each level of annuitization. Specifically, 100% annuitization has high success at the lowest withdrawal rates and drops dramatically to 0% at a withdrawal rate of about 6.5%. This is because an annuity can either pay for your monthly expenses or it cannot. Full annuitization, no matter the simulation, leads to fairly binary results. This steep drop-off becomes less dramatic as the annuity amount decreases, moving all the way to 0% annuitization.

We can see an inflection point at a withdrawal rate of about 4.8%. Roughly speaking, with a withdrawal rate below 4.8%, more annuitization results in a higher probability of success. Conversely, with a withdrawal rate above 4.8%, higher annuitization always results in a lower probability of success.

So should a retiree annuitize? If so, what level of annuitization is optimal? The decision regarding how much to annuitize depends on the level of risk a retiree feels comfortable with and their retirement goals. We attempt to address these factors through consideration of both longevity risk and the bequest goals of a retiree. Based on these results, annuitization should only be considered if a retiree's withdrawal rate is between about 3%–4.8%. Beyond a 4.8% withdrawal rate, 100% of savings should be invested in equities to maximize a retiree's probability of success. Higher stock market returns also result in much higher portfolio values at the time of death leaving the retiree with no incentive to annuitize. On the other hand, withdrawal rates lower than 3% have success rates no lower than 99.5% for all levels of annuitization. With practically no longevity risk, it would be most rational not to annuitize at all given the higher returns a stock market portfolio will produce on average.

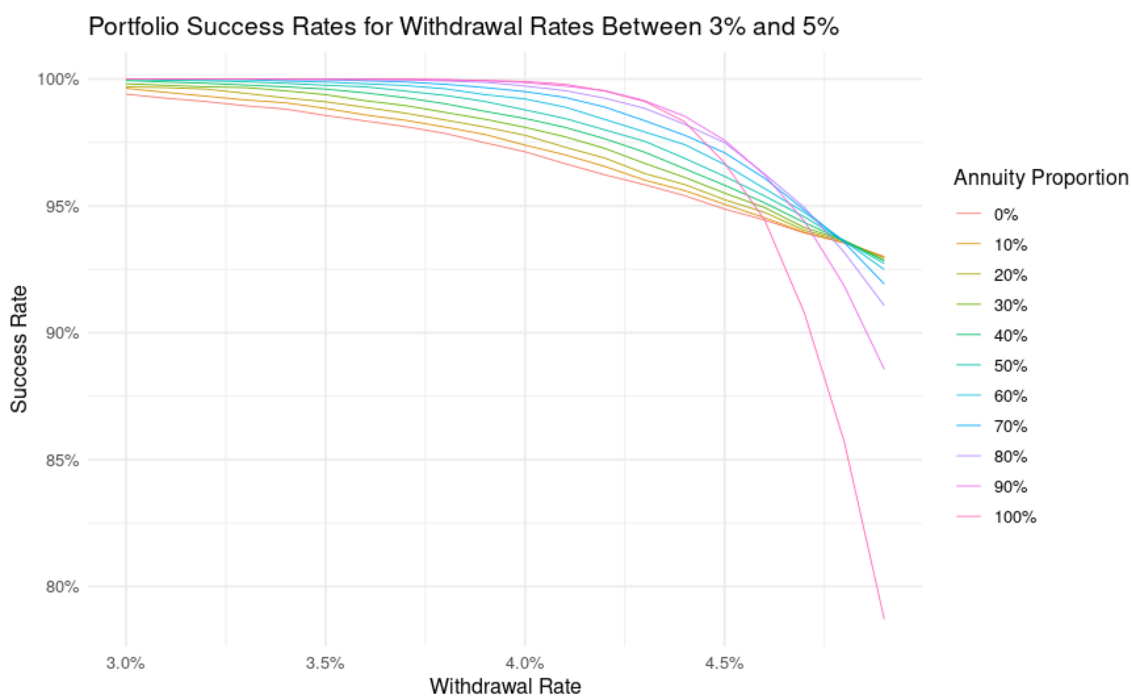

Figure 4.2: Portfolio Success Rates for Withdrawal Rates between 3% and 5%

Figure 4.2 illustrates the same success rates as in Figure 4.1, but focused only on 3%–5% withdrawal rates. Within this range of consumption rates, it could be reasonable for a retiree to consider some level of annuitization. For example, a withdrawal rate of 4.2% has success rates of 99.5% with full annuitization and 96.2% with no annuitization. An annuitization decision can reduce longevity risk by about 3%. This reduction in risk, however, comes at the cost of substantially lower bequests. The median portfolio value at death with full annuitization at a 4.2% withdrawal rate is about one-fifth the median portfolio value at death with no annuitization. So, while annuitization can provide a substantial decrease in longevity risk, the choice depends highly on retiree preferences. Retirees who place a high value on bequests may choose low levels of annuitization, accepting the increased longevity risk in exchange for a greater expected bequest. Conversely, retirees who prioritize protection against longevity risk may prefer higher levels of annuitization, accepting a much lower portfolio value at the time of death.

Commonly accepted in financial planning is the 4% rule, which states that once your yearly consumption is 4% of your retirement savings, you are safe to retire. The study that popularized this rule by Cooley et al. (1998) found using past stock market performance and inflation trends that a 4% withdrawal rate from a portfolio in 100% stocks resulted in a 95% success rate across a 30-year horizon. Our simulation results showed a 97% success rate corresponding to a portfolio with 100% equities at a 4% withdrawal rate. Aside from the use of an ESG, there are important distinctions in how these success rates are obtained. Cooley et al. (1998) deem a simulation successful only if a portfolio is not fully exhausted after a 30-year horizon. We deem a simulation successful not if a portfolio is fully exhausted at any point, but if it is fully exhausted before death. Ignoring death ages in our simulation (with a 30-year horizon) we found a success rate of 95.2%, matching closely the findings of Cooley et al. (1998). Although there is criticism of the 4% rule as an over-conservative benchmark, evaluating it under stochastic draws using the ESG provides a verification for the methods behind its creation. However, taking into account a simple additional factor like death ages highlights the over-conservative nature of the rule. With this ESG we can implement additional sources of uncertainty that highlight the randomness of economies and life events, and further get more complete answers to retirement questions. In this way, the ESG provides the flexibility of analyzing more complicated, and perhaps more realistic, retirement scenarios within a unified stochastic framework.

5 Conclusion

In this project, we introduce a new ESG built from well-known models and ESG structures for the purpose of evaluating the risk associated with retirement plans. We then use simulated draws from the ESG to better understand what levels of spending in retirement are sustainable and whether or not a retiree should annuitize.

We find that given a withdrawal rate above 4.8% and below 3%, it is in the retiree's best interest not to annuitize any of their savings. With a withdrawal rate higher than 4.8%, sustaining withdrawals through retirement depends on achieving portfolio growth that exceeds the effective interest rate built into annuity pricing. The stock market's returns on average are higher than going interest rates, and those higher returns are necessary for a retiree to avoid ruin. At withdrawal rates below 3%, longevity risk is minimal, so annuitization offers little additional protection at the cost of lower returns and is therefore unlikely to be optimal regardless of a retiree's risk tolerance. At withdrawal rates between 3% and 4.8%, careful consideration of a retiree's risk preferences and bequest goals can inform the decision on whether to annuitize.

With simulation through this ESG, many more complex and realistic financial scenarios can be easily evaluated. Rather than assuming a constant level of spending, the framework allows for testing alternative spending adjustment strategies over retirement. Complex dynamics relating to taxes, tax-benefit retirement accounts, health insurance in retirement, and end-of-life expenses can all be integrated into simulations to provide more realistic assessments of longevity risk associated with a retirement plan. Wage inflation and bond returns are the pieces of our proposed ESG in figure 3.1 that weren't implemented in the analysis presented here. Along with bond returns, we can explore different investment opportunities retirees have outside of a broad-based index fund and a life annuity (actively managed mutual funds, CDs, etc.). Further, this framework can be used to assess an individual's financial future before retirement. With wage inflation, we can properly incorporate income adjustments into the simulation as other economic factors change, and extend scenarios beyond just retirement.

With a tool that provides stochastic draws of the future economy, financial plans can be evaluated not only based on past scenarios but across a wide range of potential future scenarios. By more thoroughly exploring the space of possible economic scenarios, this framework provides greater confidence in the assessment of a retirement plan's sustainability, and as financial markets, public policy, and individual circumstances evolve, this type of simulation-based approach offers a flexible method for evaluating retirement strategies and supporting more informed decision-making.

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